

YIWEI YANG

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EDUCATION

UC Santa Cruz, Ph.D. Student 08/2022 – 06/2028

- Major: Computer Science, advised by Andi Quinn. 22 Fall TA of Computer Architecture

ShanghaiTech University, Undergraduate 09/2018 – 06/2022

- Major: Computer Science, finished Compiler, Network, Database, OS, CA, Convex, RL, Parallel Computing. 21& 22 Spring TA of Compiler

WORK EXPERIENCE

Oxmiq Labs, Campbell, CA 06/2025 – 09/2025

(Compiler Engineer) Compiler Engineer Intern

- Applied HetGPU into Production.

Jump Trading, Shanghai, China 07/2020 – 09/2020

(Linux Team) Production Engineer Intern

- High Frequency Trade Order Book simulation applying Linear Regression Method.
- Designed a user interface to automate the core affinity of jobs.
- Designed an eBPF exporter of GPFs full OSS lifetime traces for better reporting the bottleneck.

RESEARCH EXPERIENCE

Storage Systems Research Center, UC Santa Cruz 08/2022 – Present

(Graduate Research) Assistant

- Data-driven CXL.mem allocation, prefetching, and replacement policies Operating System. 🏆
- Make Hardware Software Co-design for on CXL.cache data movement 📦
- Make universal migration based on WebAssembly and PTX^{MV}_{VM}
- Make cross kernel-userspace eBPF observation for distributed system 🐝
- Contributed to vLLM/SGLang for high-throughput serving, implemented custom CUDA CXL kernels for attention mechanisms, optimized KV cache management and continuous batching strategies for improved GPU utilization. 🧩

SELECTED PUBLICATIONS

”**SlugArch: A Boundary Replay Contract for Heterogeneous CXL Systems**” Preprint **Yiwei Yang**, Andi Quinn.

”**WIO: Upload-Enabled Computational Storage on CXL SSDs**” Preprint **Yiwei Yang**, Yanpeng Hu, Yusheng Zheng, Estabon Ramos, Jianchang Su, Andi Quinn, Wei Zhang.

”**CXLMemUring: Revolutionizing Memory Access Through Compiler-Driven Heterogeneous Execution**” Preprint **Yiwei Yang**, Yusheng Zheng, Yuyi Li, Jianchang Su, Wei Zhang, Andi Quinn.

”**MVVM: Heterogeneous Checkpoint/Restore with Host-Specialized AOT Execution**” Preprint **Yiwei Yang**, Aibo Hu, Yusheng Zheng, Brian Zhao, Xinqi Zhang, Andi Quinn.

”**TernIP: Scaling Ternary Large Language Models**” Preprint Ethan Sifferman, Eddy Abban, **Yiwei Yang**, Dustin Richmond.

”**Concordia: JIT-Compiled Persistent-Kernel Checkpointing for Fault-Tolerant LLM Inference**” Preprint Yuhang Gan, **Yiwei Yang**, Yuyi Li, Xiangyu Gao, Yichen Wang, Rain Jiang, Xiaoning Ding, Andi Quinn, Chen Qian.

”**Safe and Programmable OS-Level GPU Resource Management with eBPF**” ASPLOS27 Yusheng Zheng, Tong Yu, **Yiwei Yang**, Minghui Jiang, Xiangyu Gao, Jianchang Su, Yanpeng Hu, Wenan Mao, Wei Zhang, Dan Williams, Andi Quinn.

”**OCEAN: Open-source CXL Emulation at Hyperscale Architecture and Networking**” SC26 **Yiwei Yang**, Mujahid Al Rafi, Kenza Bouasker, Zhen Peng, Xi Wang, Jesun Sahariar Firoz, Sayan Ghosh, Daniel Wong, Dong Li, Hyeran Jeon, Kevin Barker, Nathan Tallent, Andres Marquez and Luanzheng Guo.

”**hetGPU: Pushing pooling into heterogeneous GPU Domain**” HCDS26 **Yiwei Yang**, Yusheng Zheng, Tong Yu, Andi Quinn.

”**CXLMemSim: Practical Performance Simulation and Characterization of CXL 3.0 Memory Systems**” HPDC26 **Yiwei Yang**, Pooneh Safayenikoo, Jiacheng Ma, Tanvir Ahmed Khan, Andi Quinn.

”**Extending Applications Safely and Efficiently**” OSDI25 Yusheng Zheng, Tong Yu, **Yiwei Yang**, Yanpeng Hu, Xiaozheng Lai, Dan Williams, Andi Quinn.

”**Kgent: Kernel Extensions Large Language Model Agent**” eBPF24 Yusheng Zheng, **Yiwei Yang**, Maolin Chen, Andi Quinn.

SKILLS

- **Programming Languages:** not limited to any specific language, and experienced in Verilog/C++/Rust, comfortable with Golang/C/Java/TypeScript (in random order).
- **System:** Specialist in Compiler & Performance Analysis, familiar with LLVM, MLIR, Gem5, WASM, gdb, eBPF, ptx, qemu, chisel, Linux mm & observability subsystem.
- **Machine Learning:** familiar with general knowledge of machine & reinforce learning, interested of System for LLM for Sys.

MISCELLANEOUS

- Interests: HPC, Architecture, Security, JIT, CXL, LLM4Sys4LLM, etc.
- Lead GeekPie_HPC *Ranked 2*, SC-SCC21. *Ranked 4*, ISC22. Advise Not-Slow-Slug *Ranked 2*, ISC23.
- Advise Kiki Zhang, Zhen Cao to be Ph.D. students.